

Methodology to Predict Random Telegraph Noise Induced Threshold Voltage Shift Using Machine Learning

Eunseok Oh*, Jang Kyu Lee, Youngsoo Seo, and Hyungcheol Shin

Inter-University Semiconductor Research Center (ISRC) and School of Electrical Engineering and Computer Science, Seoul National University, Seoul 151-747, Korea

e-mail: bakhacandy@snu.ac.kr; phone: +82-2-880-7279

Abstract—We suggest the methodology to predict the distribution of threshold voltage (V_t) shift caused by random telegraph noise (RTN). Poly-silicon channels were randomized with a single trap and the neural network was modeled to predict RTN trap-induced V_t fluctuation in 3D NAND Flash Memory. 3D Technology Computer-Aided Design (TCAD) simulations were performed in a unit cell to calculate the V_t shift in a 3D vertical channel. Finally, we extract the distribution of V_t fluctuation using machine learning.

Keywords—3D NAND, Random Telegraph Noise, Machine Learning, Neural Network.

I. INTRODUCTION

In 3D NAND Flash Memories, it is important for cells to maintain accurate threshold voltages (V_t) because they indicate the stored data. By various reasons, however, V_t shift can occur. The V_t fluctuation by Random Telegraph Noise (RTN) is one of them [1][2], which can cause serious failures. Recently, as poly-silicon is more widely used, importance of RTN increases because in poly-silicon channel, there exists a “Percolation Effect” [3]. To estimate the failure rate due to RTN, distribution of RTN induced V_t shift amplitude is needed. Previously, it can be obtained by numerous simulations, which leads to immeasurably long time. However, with machine learning, we obtained the distribution with far less simulations in much shorter time. There are many features that changes the amplitude of RTN, [4], [5] among which the grain boundary configuration and the location of a trap are very decisive. That’s because the area of grain boundary and the locations of trap can differentiate the number of grain boundary traps and “Percolation Effect” respectively [2], [3]. In this paper, we investigated the V_t shift according to various grain boundary configurations and locations of traps by proposing a new way to describe them logically and applying machine learning.

II. CELL STRUCTURE RANDOMIZED FOR MACHINE LEARNING

A unit cell in 3D Macaroni NAND Flash Memory is used in research. It is reasonable because threshold voltage is measured in reading process, which means other cells in the same string are “On-state”. The cell is simplified by eliminating charge trap layer and blocking oxide to focus on machine learning methodology. In all devices, the basic geometry are equal. The filler radius, the channel thickness and the oxide thickness are kept in constant at $R_{\text{filler}} = 8$ (nm), $T_{\text{ch}} = 5$ (nm), $T_{\text{ox}} = 5$ (nm) respectively. Word Line (WL) Length and Spacer Length are also kept in constant at $L_{\text{WL}} = L_{\text{spacer}} = 30$ (nm). Fig 1 (a) and (b) show the 3D structure of the unit cell and cross section perpendicular to X-axis. 0.7V is applied to the drain of the unit cell, and WL voltage was swept from 0V to 1V. The TCAD (Technology Computer-Aided Design) simulations are performed by Sentaurus [6].

To apply machine learning, there should be appropriate inputs that have direct effect on outputs. In this paper, two are selected; Grain boundary location and trap location. Fig. 2 shows enlarged view of part “A” in Fig. 1 with how the grain boundary configuration is randomized. Regarding average grain boundary size is 30nm, the channel is assumed to be sliced once vertical to Z-axis. “Grain Boundary Location” represents the distance from source to the middle point of the grain boundary. The location of a trap is randomized easily by its coordinates. The coordinates of the trap are assumed to be on the grain boundary, which has most impact on RTN trap-induced V_t shift. As the channel is cylindrical symmetric, trap is also assumed to be located on $y=0$ plane. So, “Trap Location” represents the distance from center of the filler to the RTN trap. Fig. 3 shows 3D structure of sliced channel when Grain Boundary Location = 16.753 (nm), representing poly-silicon channel.

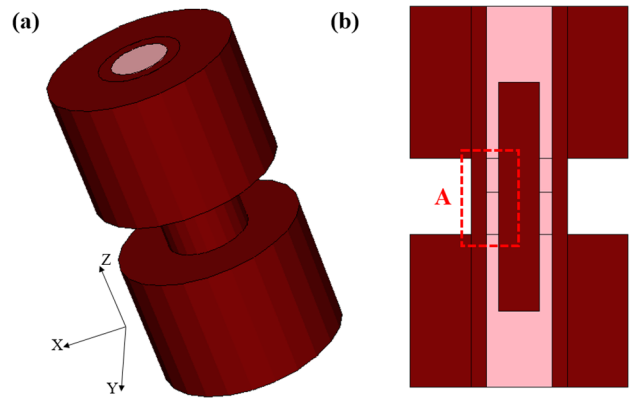


Fig. 1. (a) 3D structure of the unit cell (b) cross section of the cell perpendicular to X-axis.

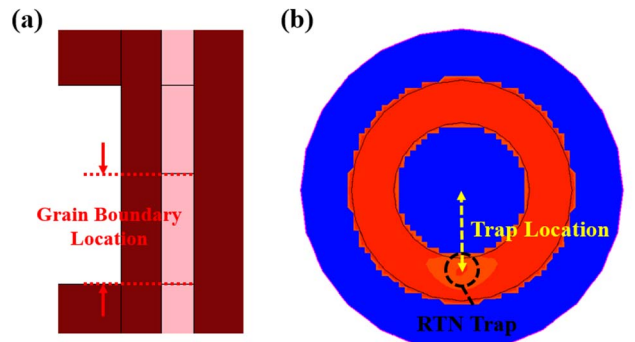


Fig. 2. (a) Enlarged view of part “A” and randomized grain boundary (b) cross section of the cell perpendicular to Z-axis with randomized trap.

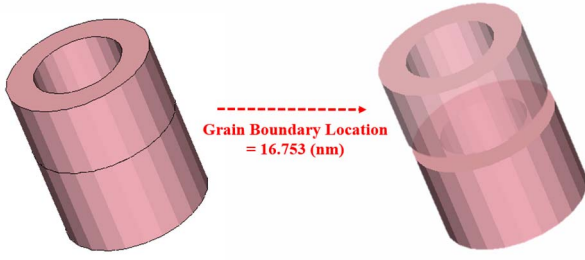


Fig. 3. Poly-silicon channel structure when Grain Boundary Location value is 16.753 (nm).

The amplitude of V_t shift by RTN in each case is obtained by subtracting threshold voltages when an electron is trapped and detrapped. Fig. 4 shows the current density in trapped case is remarkably low around the trap, causing the higher threshold voltage. As threshold voltages are extracted by constant current method (10nA), V_t shift can be seen more clearly in I-V curves shown in Fig. 5.

Machine learning is consisted of two main processes; Training and Test. A neural network model trained itself during training to decrease Mean Square Error (MSE) between simulations and its predictions by mathematical model “Gradient Descent”[7]. Then the neural network is tested how well it is trained. Thus TCAD simulations were also divided into two sets. Some of data are training sets used in training process, on the other hand, the rest of the data are test sets used in test process. The neural network is designed with 2 inputs of “Grain Boundary Location” and “Trap Location”. The outputs are the values of V_t shift, V_t when an electron is trapped, and detrapped that count in calculating amplitudes of V_t shift. The final neural network model is shown in Fig. 6. The network has 3 Hidden Layers with 100 nodes each.

Before training the network, all the data should be preprocessed, which means all the units and scales have to be matched and optimized.

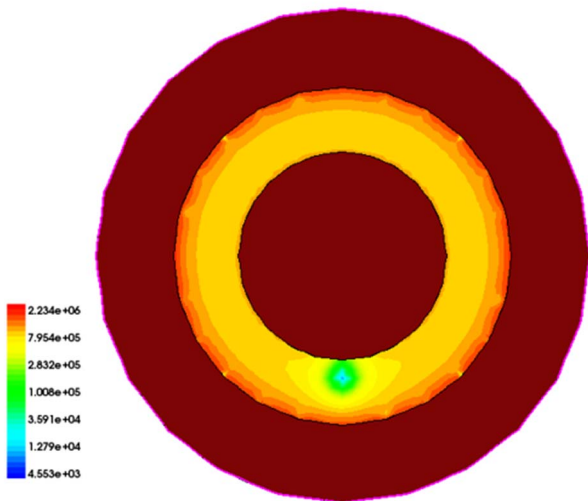


Fig. 4. Current density at Z = Grain Boundary Location. The same cell with figures above.

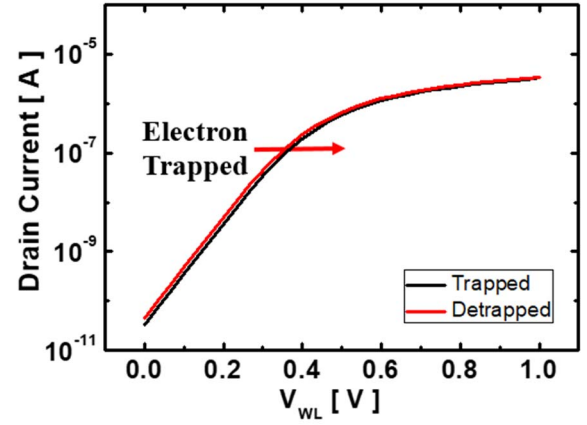


Fig. 5. Drain current – WL Voltage curve shifts to right when electron is trapped. Accordingly, the V_t increases.

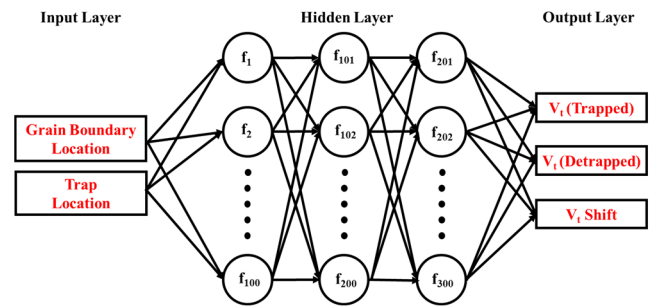


Fig. 6. The Final Neural Network Model

III. RESULTS

2000 simulations were run, the half of which were used for training and the remaining data for test. As the target error is set to 1%, the training proceeded until the network reaches the target. Fig. 7 shows the error rate becomes smaller over time, as the network trained.

Histograms in Fig. 8 (a), (b), and (c) show the results of training process, while Fig. 8 (d), (e), and (f) show that of test process. Almost same numbers of cells are in the same range. Exact error rates are presented in Table 1, representing the high accuracy of the network obviously.

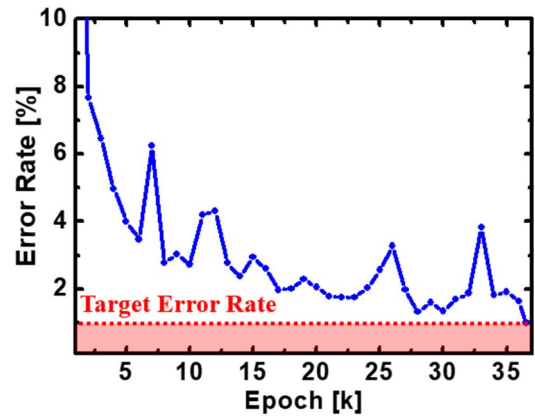


Fig. 7. Error rate becomes smaller over time until the network reaches the target error rate.

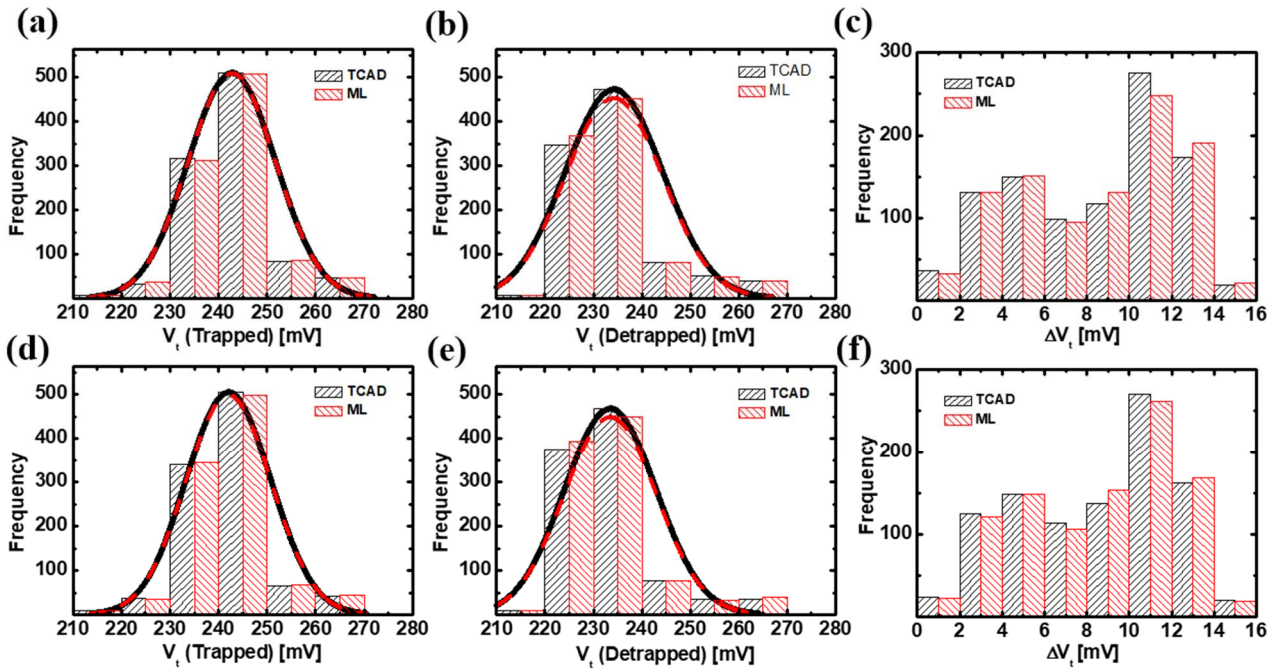


Fig. 8. (a), (b) and (c) represents results of V_t (Trapped), V_t (Detrapped) and V_t Shift (ΔV_t) predictions with the TCAD simulation data of training sets respectively. (d), (e) and (f) represents results of V_t (Trapped), V_t (Detrapped) and V_t Shift (ΔV_t) predictions with the TCAD simulation data of test sets respectively.

TABLE I. NEURAL NETWORK ERROR RATE OF TEST PREDICTIONS

1000 strings (Test Set)	Parameters				
	Mean, V_t (Trapped)	Mean, V_t (Detrapped)	Standard Variation, V_t (Trapped)	Standard Variation, V_t (Detrapped)	ΔV_t
Error Rate	0.015%	0.015%	0.29%	0.068%	3.27%

Fig. 9 indicates the high accuracy by comparing true simulation values and neural network predictions. The network is more precise, as symbols are closer to the red line.

The cumulative distributions of the V_t shift predictions are in Fig. 10, showing the high accuracy of predictions again. From these results, we can assume that the network is trained well enough to predict the amplitudes of V_t shift by RTN in random cells.

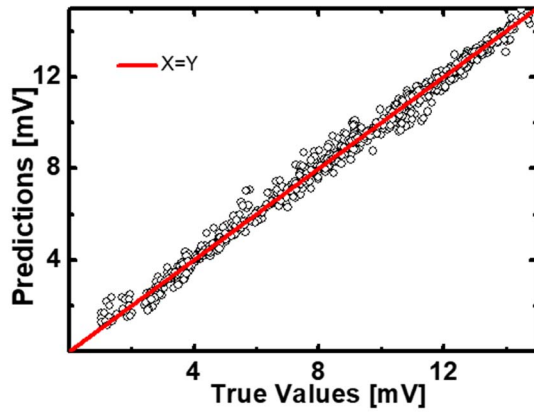


Fig. 9. Comparing true simulation data and prediction results of the network. As the red line represent $X=Y$, the symbol would be close to it if prediction is precise.

So if enough numbers of random values for each inputs are extracted and put into the network, the representative distributions for all cells can be extracted. 30000 sets of inputs are obtained randomly and predicted by the trained network. Fig. 11 shows the cumulative distributions of random cells.

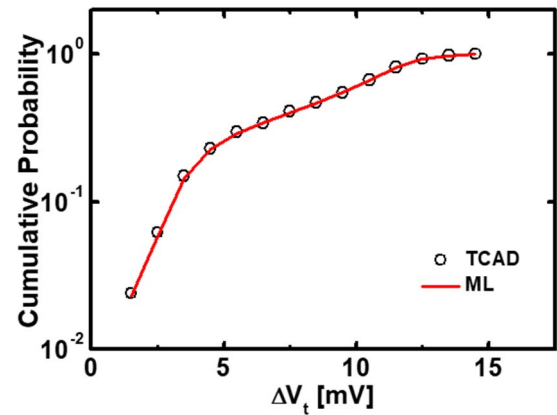


Fig. 10. (a) and (b) represent the cumulative distribution of predicted ΔV_t of training and test sets respectively.

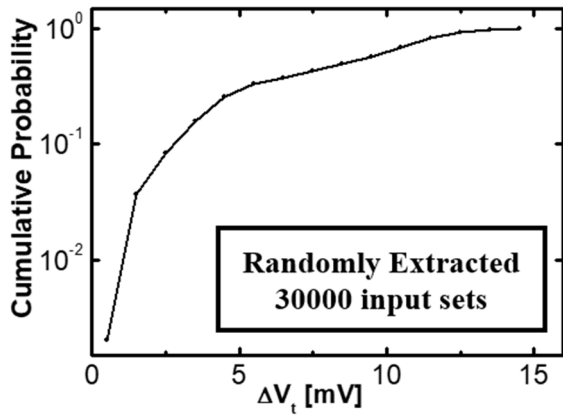


Fig. 11. The cumulative distribution of predicted ΔV_t for randomly extracted 30000 input sets (Grain Boundary Location, Trap Location).

It is also possible to predict various tendencies without splitting the parameters. We made input sets of random grain boundary location with fixed trap location and of random trap location with fixed grain boundary location. Fig. 12 shows the result when former is put into the network predicting the tendency when trap is located in certain position, as the grain boundary moves from source to drain. Fig. 13 shows the amplitude of V_t shift at certain grain boundary location, as the trap moves outward from the filler/channel interface to channel/oxide interface.

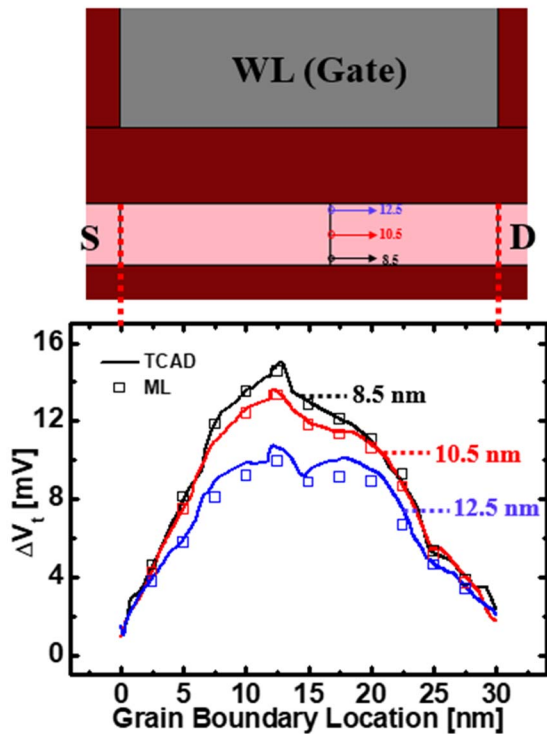


Fig. 12. Simulation data (Line) and predictions (Symbol) when trap location value is 8.5 nm (black), 10.5 nm (red), 12.5 nm (blue) as the grain location value increases from 0 nm to 30 nm.

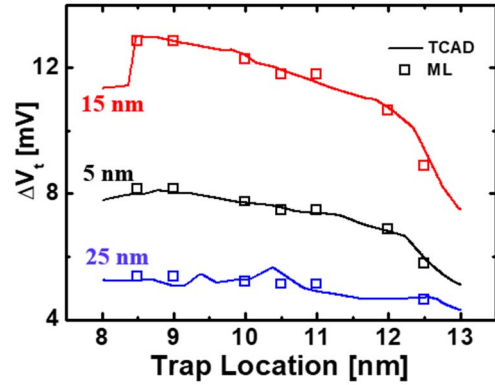


Fig. 13. Simulation data (Line) and predictions (Symbol) when grain boundary location value is 5 nm (black), 15 nm (red), 25 nm (blue) as the trap location value increases from 8 nm to 13 nm.

IV. CONCLUSIONS

In this paper, the new methodology for predicting amplitudes and extracting distributions of V_t shift by RTN was suggested. 2000 cells with random transverse grain boundary with randomly located traps were simulated. Using those data, the neural network was trained and predicted the amplitudes of V_t shift by RTN. The high accuracy of the network was also shown. Moreover, the network predicted the amplitudes of V_t shift for completely new 30000 input sets extracted aimlessly by random grain boundary and trap generator. Finally, the full distribution of V_t shift was obtained. It implies that with only 2000 TCAD simulations, results of numerous simulation can be predicted, leading to much shorter time to get data. The network once modeled is also useful to extract tendencies under certain situations.

ACKNOWLEDGMENT

The authors would like to thank Synopsys for providing TCAD tool.

REFERENCES

- [1] D. Resnati, A. Mannara, G. Nicosia, G. M. Paolucci, P. Tessariol, A. S. Spinelli, A. L. Lacaita, and C. M. Compagnoni, "Characterization and Modeling of Temperature Effects in 3-D NAND Flash Arrays—Part I: Polysilicon-Induced Variability," *IEEE Transactions on Electron Devices*, vol. 65, no. 8, pp. 3199–3206, August 2018.
- [2] D. Resnati, A. Mannara, G. Nicosia, G. M. Paolucci, P. Tessariol, A. S. Spinelli, A. L. Lacaita, and C. M. Compagnoni, "Characterization and Modeling of Temperature Effects in 3-D NAND Flash Arrays—Part II: Random Telegraph Noise," in *IEEE Transactions on Electron Devices*, vol. 65, no. 8, pp. 3207–3213, Aug. 2018.
- [3] Y. L. Chou, Tahui Wang, Mercator Lin, Y. W. Chang, Lennis Liu, S. W. Huang, W. J. Tsai, T. C. Lu, K. C. Chen, and Chih-Yuan Lu, "Poly-Silicon Trap Position and Pass Voltage Effects on RTN Amplitude in a Vertical NAND Flash Cell String," in *IEEE Electron Device Letters*, vol. 37, no. 8, pp. 998–1001, Aug. 2016.
- [4] D. Resnati, A. Mannara, G. Nicosia, G. M. Paolucci, P. Tessariol, A. S. Spinelli, A. Goda, A. L. Lacaita, and C. M. Compagnoni, "Impact of temperature on the amplitude of RTN fluctuations in 3-D NAND flash cells," *2017 IEEE International Electron Devices Meeting (IEDM)*, San Francisco, CA, 2017, pp. 21.3.1–21.3.4.
- [5] Gianluca Nicosia, Akira Goda, Alessandro S. Spinelli, Christian Monzio Compagnoni, "Impact of Cycling on Random Telegraph Noise in 3-D NAND Flash Arrays," in *IEEE Electron Device Letters*, vol. 39, no. 8, pp. 1175–1178, Aug. 2018.
- [6] *Sentaurus Device User Guide*, Synopsys, Zurich, Switzerland, 2016.
- [7] Dmitry Yarotsky, "Error bounds for approximations with deep ReLU networks" *Neural Networks*, vol. 94, pp. 103–114, October 2017.